Environmental Product Declarations in Construction: Mineral Wool vs. XPS Insulation

Which insulation supports a more sustainable decision?

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Products and their EPDs

- **Product A:** ROCKWOOL *Safe'n'Sound* stone wool batt, ROCKWOOL North America
- **Product B:** URSA XPS NVII L extruded polystyrene board, URSA Insulation S.A. (Madrid), Bondeno plant (Italy)
- **Functional unit:** 1 m^2 insulation at a declared thermal resistance — not the same value (A: RSI 1.0; B: $R = 1.50$ at 50 mm); normalised next slide
- **EPD programme:** A: UL Environment (ISO 14025), EPD 4789092768.101.1; B: International EPD® System S-P-09927 (EN 15804+A2), ver. M. Gómez Ferrer
- **System boundary:** cradle-to-grave in both (A: A1–C4; B: A1–A3, A4, A5, B1–B7, C1–C4, plus D)





Product images: ROCKWOOL *Safe'n'Sound* (left) and URSA XPS NVII L (right).

Environmental performance comparison

Indicator (per 1 m^2 of insulation)	Product A (RSI 1.0)	Product B ($R = 1.50$)
GWP-total, A1–A3, as declared	1.31 kg CO_2e	7.01 kg CO_2e
GWP-total, A1–A3, scaled to $R = 1.50$	≈1.97 kg CO_2e	7.01 kg CO_2e
GWP-total, A1–C4 (cradle-to-grave)	2.03 kg CO_2e	7.75 kg CO_2e
Total non-renewable primary energy, A1–A3	n.d.	136 MJ
Net fresh water use, A1–A3	n.d.	0.119 m^3

Table 1: Declared environmental indicators per functional unit of insulation.

Sources: URSA EPD S-P-09927 (A1–A3, 50 mm, $R = 1.50$); ROCKWOOL GWP look-up tables (RSI 1.0). A1–C4 = A1–A3 + A4 + A5 + C2 + C4. n.d. = not declared; scaling to $R = 1.50$ assumes proportionality (indicative). BEAM reference, kg CO_2e per m^2 RSI: mineral wool 3.30, XPS HFC-blown 55.50, low-GWP 4.90.

Interpreting the results

- **GWP:** mineral wool wins on every basis — 1.97 vs. 7.01 kg CO_2e per m^2 at $R = 1.50$; XPS carries $\approx$3.6$\times$ the product-stage footprint
- **Energy and water:** ranking cannot be checked — ROCKWOOL's tables publish only GWP, so XPS's 136 MJ and 0.119 m^3 cannot be benchmarked yet
- **Magnitude of the gap:** real, but far below the $>$10$\times$ quoted for HFC-blown XPS (55.50 vs. 3.30 per RSI); 7.01 indicates a low-GWP blowing agent (type undisclosed)
- **Verdict:** no unconditional winner — mineral wool is preferable on GWP for interior walls, but other indicators and the application still matter
- **Caveats:** functional units differ (RSI 1.0 vs. $R = 1.50$), so the scaled row is indicative; different programmes and PCRs limit strict comparability

Comparison and decision

- **Recommendation:** specify mineral wool for low-carbon interior partitions; if XPS is required (inverted roofs, below-grade, load-bearing), demand blowing-agent disclosure and prefer low-GWP variants
- **Environmental trade-off:** XPS trades a higher GWP for CS(10/Y)700 compressive strength (700 kPa), moisture resistance and a 50-year service life — properties mineral wool cannot match structurally
- **Space-efficiency trade-off:** XPS reaches the same R-value at $\lambda \approx 0.034$–0.036 W/(m·K) in a thinner profile; mineral wool needs more build-up, a hidden cost of its lower GWP
- **Most interesting finding:** "XPS $=$ high carbon" is outdated — low-GWP agents shrink the gap from $>10\times$ to $\approx 3.6\times$; category labels alone cannot drive specification
- **Information still needed:** exact blowing agent; ROCKWOOL primary energy and water; A4 transport distance; fire class and end-of-life — one indicator is not the whole story

References

- **EPD, Product B:** URSA Insulation S.A., *URSA XPS NVII L*, International EPD System, reg. S-P-09927 / EPD-IES-0009927:002, 2023 – https://www.environdec.com/library/epd9927
- **EPD, Product A:** ROCKWOOL, *Safe'n'Sound*, UL Environment EPD no. 4789092768.101.1, GWP look-up tables – https://www.rockwool.com/siteassets/o2-rockwool/documentation/technical-bulletins/commercial/epd-gwp-look-up-tables.pdf
- **Database:** One Click LCA, *Stone wool insulation batt, Safe n Sound, Rockwool North America* – https://materials.oneclicklca.com/en/material/stone-wool-insulation-batt-for-interior-walls/62d145282102db6d5dadbbdd
- **Standards:** EN 15804:2012+A2:2019, ISO 14025:2006; PCR 2019:14 v1.11 and C-PCR-005 (EN 16783)
- **Background data:** Vermont Public Service Department, *Default Insulation GWP Values* (BEAM); EPDs accessed 23 October 2026